5. We then repeat this process one hundred times, each time using the posterior probability of the previous iteration as our prior for the current iteration.

What we need to know on a high level here is that after each test, our posterior belief that this person is a psychic *never decreases*, since that statement doesn't agree with the data we observe. Specifically, Figure 3.2 shows the progressive posterior probability of each person in our group as a function of how many tests the woman at the carnival has passed.

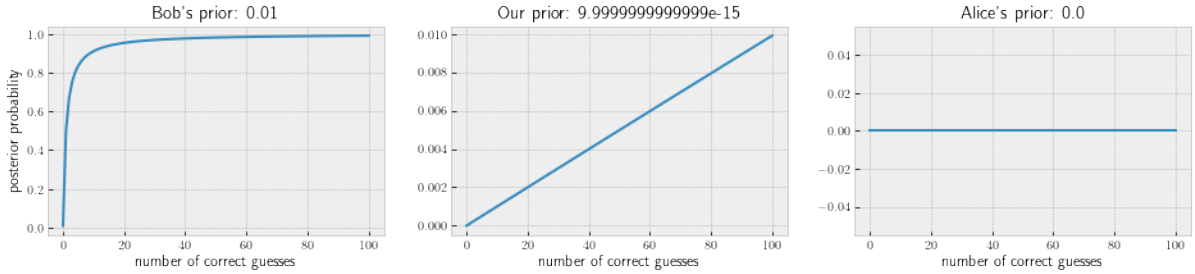


Figure 3.2 Progressive posterior probability that the woman at the carnival is a psychic as a function of the number of successful guesses. This posterior never decreases, but behaves differently depending on the initial prior.

As we can see, each of the three curves either increases or stays flat—none of them actually decreases, since a decreasing probability of being a psychic doesn't match the one hundred successful guesses in a row. But why do the three curves look so different? As you might have already guessed, the starting position of the curve, that is, each person's prior probability that the woman is a psychic, is the cause.

In Bob's case in the left panel, he started out with a relatively high prior that the person is a psychic, 1 percent—Bob is a believer. As he observes more and more data that agree with this belief, his posterior probability increases higher and higher.

In our own case in the middle, being a skeptic, we started with a much lower prior, 1 in 10 raised to the 14th power. However, since our observations do suggest that the woman is a psychic, our posterior probability also increases as more data come in, reaching 1 percent at the end.

Alice's case on the right, on the other hand, is different. From the start, she didn't believe that psychics are real, so she assigned exactly zero probability to her prior. Now, remember that according to Bayes' theorem, the posterior probability is proportional to the prior multiplied by the likelihood. Since Alice's prior is exactly zero, this multiplication in the Bayesian update will always produce another zero.

Since Alice started out with zero probability, even after a successful test, this probability only stays the same. After one correct guess, Alice's posterior is zero. After two guesses, it's still zero. After all one hundred correct guesses, this number is still zero. Everything is consistent with the Bayesian update rule, but because Alice's prior doesn't allow for the possibility that psychics exist, no amount of data could convince her otherwise.

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